Testing Plan for Interactive Prototype 1: XR Event Planning

3-Sentence Pitch

XR Event Planning is a VR event planning application that allows users to arrange and customize an event venue from inside the space. Users can place and manipulate furniture, adjust the venue lighting, and change the overall theme using physical XR interactions. The aim is to make venue planning more direct and spatial compared with using a traditional 2D interface.

Testing Objective

From my concept, I identified an assumption that users would be able to understand the main physical XR interactions for arranging and customising an event venue without being told exactly how each interaction works. This test aims to evaluate whether participants can independently:

- Arrange furniture within the venue
- Identify how unwanted furniture can be removed
- Adjust the venue lighting
- Change the venue theme

The testing will also identify where participants hesitate, make incorrect attempts, or require assistance. These observations will help identify which interactions are understandable and which need to be refined in the next iteration.

Testing Methodologies

This testing session will use task-based usability testing, observation, and think-aloud feedback to evaluate the Unity prototype. Task-based usability testing is appropriate because the main assumption concerns whether users can discover and complete the prototype's key interactions without step-by-step instructions. Participants will therefore be given goals rather than being told which interface elements to use.

During each task, I will observe the participant's first actions, moments of hesitation, incorrect attempts, repeated mistakes, and whether assistance is required. Participants will also be encouraged to think aloud while completing the tasks. This will provide additional insight into what they expect the interactions to do and why particular parts of the prototype may feel clear or confusing.

Using these methods together will provide both task-completion data and qualitative feedback that can be used to identify specific improvements for the next prototype iteration.

Prototype Description / Requirements

The prototype is a simplified VR event venue designed to test the main interactions for arranging and customising an event space. The prototype currently allows users to:

- Add tables, chairs, and a stage to the venue
- Grab and move furniture around the space
- Rotate furniture
- Remove furniture by physically moving it into a red Delete Zone
- Adjust the venue brightness by physically moving a lighting lever
- Change between Warm, Modern, and Dark venue themes by grabbing the corresponding wand and placing it into a pedestal
- Remove a wand from the pedestal and select a different theme
- Return wands to their original position when they are released outside the pedestal

The prototype also contains duplicate and guest review functions. These functions will not be included in this testing session because the testing will focus on the main physical XR interactions of furniture arrangement, lighting adjustment, and theme customisation.

Data Collection Method

Each participant will be identified anonymously using participant labels such as P1, P2, P3, P4, and P5. For each main task, I will record whether the participant:

- Completes the task independently
- Requires assistance
- Cannot complete the task

I will also record:

- The participant's first action
- Moments of hesitation or confusion
- Incorrect attempts or repeated mistakes
- Difficulties using individual interactions
- Comments made during the think-aloud process
- Short feedback provided after completing the prototype

Task Completion Rating

Pass (2): The participant completes the task independently.

Assist (1): The participant completes the task after receiving a hint or assistance.

Fail (0): The participant cannot complete the task even after assistance.

These ratings will allow task performance to be compared across participants while the observation notes provide additional information about why difficulties occurred.

Participant Results Table

Participant	Furniture Arrangement	Lighting Adjustment	Theme Customisation	Main Observation / Comment
P1				
P2				
P3				
P4				
P5				

For Furniture Arrangement, additional notes will be taken about adding furniture, grabbing and moving objects, rotation, and removing furniture.

Testing Setup

Before each participant begins, I will reset the Unity prototype to the same starting state. The starting setup will have:

- No newly added furniture in the venue
- Lighting at its normal starting brightness
- All three theme wands returned to their original positions
- An empty theme pedestal
- The Delete Zone, lighting lever, and theme station visible and functional

Before beginning the tasks, I will briefly explain only the basic controls required to operate the XR Interaction Simulator, such as movement, selecting a controller, and grabbing. I will not explain:

- What the Delete Zone does
- How the lighting lever works
- What the theme wands represent
- How the pedestal works

This is because operating the XR Simulator is part of the testing setup, while discovering how the prototype interactions work is part of the testing objective. Each testing session will take approximately 5 to 6 minutes.

Moderator Assistance Rules

To keep the testing process consistent across participants, the following assistance procedure will be used. Participants will first be given an opportunity to complete each task independently. If a participant becomes stuck, I will first use a neutral prompt such as: *"What are you trying to do?"* or *"What would you try next?"* If the participant is still unable to continue, I may provide a small hint without immediately explaining the exact interaction. A direct instruction will only be given if necessary for the participant to continue with the testing session. If a participant requires a hint or direct instruction to complete a task, that task will be recorded as Assist (1). I will not correct participants' opinions or explain the intended design while they are attempting a task.

Testing Process

1. Introduction and Controller Briefing

Approximately 45 seconds to 1 minute

I will introduce the session by saying:

"Hi, I'm testing an early VR prototype for planning and customising an event venue. I'm testing the prototype, not you, so there are no wrong answers. I'll give you a few goals to complete, but I won't explain exactly how to complete them because I want to see what feels understandable. Please say what you are thinking while you use the prototype, especially if something is confusing."

I will then ask:

"Are you happy to participate in this short usability test and for me to take notes on your responses?"

If the session is being recorded, I will separately ask:

"Is it also okay if I record this session so I can review your feedback later?"

I will then briefly explain the basic XR Simulator controls required to move around and operate the virtual controller. I will not demonstrate how to complete any of the prototype tasks.

2. Task 1: Arrange the Venue

Approximately 1 minute 30 seconds

I will tell the participant:

"Imagine you are preparing this venue for an event. Add some furniture and create a simple layout. Reposition the furniture, change the orientation of at least one object, and remove one object that you no longer want."

I will not tell the participant to use the Delete Zone unless assistance becomes necessary. I will observe whether the participant can independently:

- Add furniture

- Grab and move furniture
- Change the orientation of furniture
- Identify how unwanted furniture can be removed

I will also record hesitation, incorrect attempts, repeated actions, and any assistance required.

3. Task 2: Adjust the Lighting

Approximately 45 seconds

I will tell the participant:

"Now adjust the lighting so that the venue becomes noticeably darker or brighter."

I will not mention the lighting lever. I will observe:

- What the participant looks for first
- Whether they notice the physical lighting control
- Whether they understand how to use the lever
- Whether assistance is required

4. Task 3: Change the Venue Theme

Approximately 1 minute

I will tell the participant:

"Now change the overall theme of the venue."

I will not mention the wands or pedestal. I will observe:

- Whether the participant connects the wands with different themes
- Whether they understand that a wand needs to interact with the pedestal
- Any hesitation or incorrect attempts
- Whether assistance is required

If the participant completes the task quickly, I will then say:

"Now change it to a different theme."

This will test whether the participant also understands how to remove the current wand and replace it with another theme.

5. Short Post-Test Feedback

Approximately 1 minute

After the participant has completed the tasks, I will ask the same three questions:

"Which interaction felt the most natural?"

"Was there anything you did not understand straight away?"

"What is one thing you would improve?"

Any additional comments made by the participant will also be recorded.

Expected Outcome of the Testing

The testing should identify which of the three main interaction areas are understandable without detailed instructions and which require additional guidance or refinement. The combination of task completion ratings, observations, think-aloud comments, and post-test feedback will be used to identify patterns across participants and guide the next iteration of the prototype.